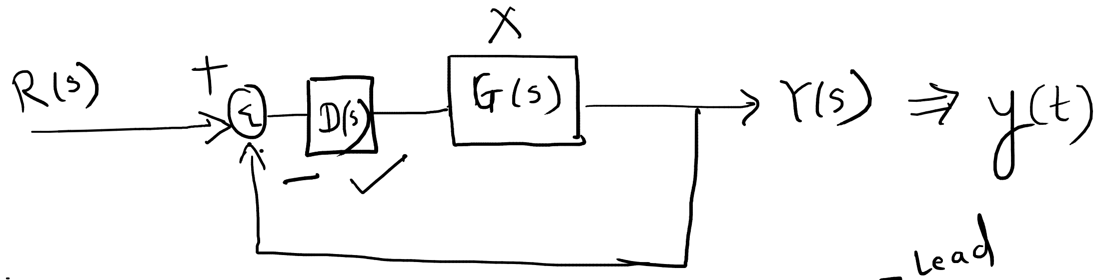
Lead - Lag

Compensator Adjust timings | Errors

ξ ω_n

t_s $\cancel{E}$ e_{ss}



D(s) → ☆ compensator → Lead / Lag / Lead-lag compensator

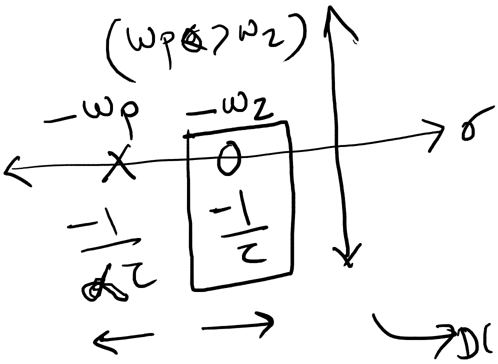
→ controller (P, I, D)

Lead - compensators D(s) ↳ extra

G(s) is already in place

$\omega_z = \omega_{c_1} = \left|\frac{-1}{\tau}\right| = \frac{1}{\tau}$ ✓

$\omega_p = \omega_{c_2} = \left|\frac{-1}{\alpha\tau}\right| = \frac{1}{\alpha\tau}$ ✓



$$D(s) = \boxed{\frac{s+\omega_z}{s+\omega_p} \cdot K}$$

$$D(s) = \frac{s + 1/\tau}{s + \frac{1}{\alpha\tau}}$$ → zero

Pole-zero

$$\boxed{D(s) = \frac{\tau s + 1}{\alpha\tau s + 1}}$$

$$D(s) = \frac{s+\omega_z}{s+\omega_p} \times \frac{\omega_p}{\omega_z}$$

$\boxed{\omega_z < \omega_p}$

$\boxed{\tau > 0}$ $\boxed{\alpha < 1}$ ✓

Time-constant

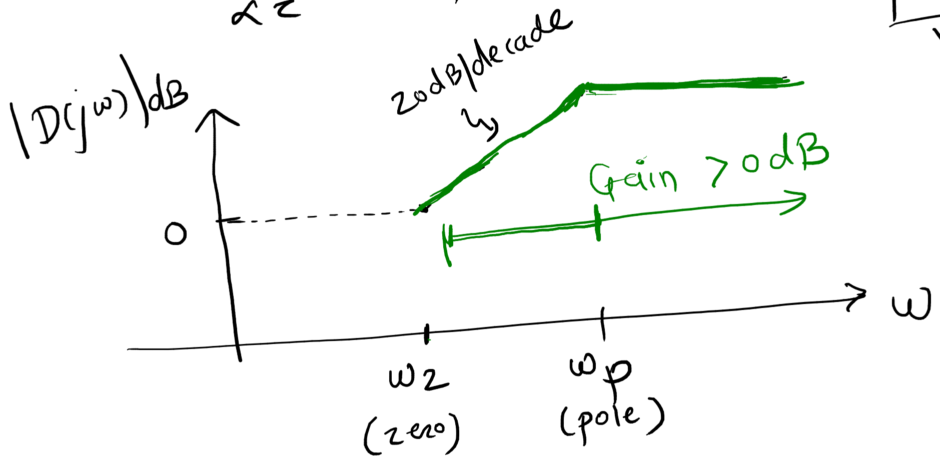
$$D(s) = \frac{s+\omega_z}{s+\omega_p} \times \frac{\omega_p}{\omega_z}$$

$$= \frac{\frac{1}{\omega_z}s+1}{\frac{1}{\omega_p}s+1} = \underline{\underline{\frac{\tau s+1}{\alpha\tau s+1}}}$$

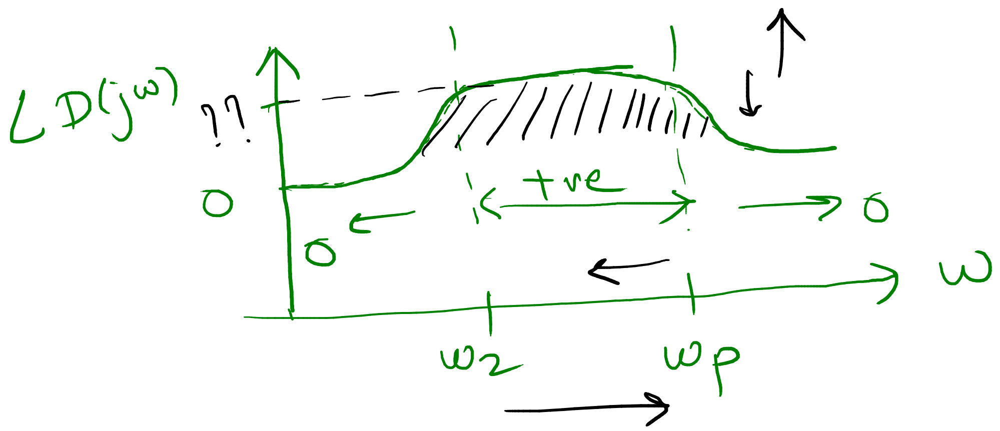
$$\boxed{\begin{aligned}\omega_z &= \frac{1}{\tau}\\ \omega_p &= \frac{1}{\alpha\tau}\end{aligned}}$$

$$\left.\begin{aligned} \omega_{c_1} &= \frac{1}{\tau} = \omega_z \quad (\text{zero})\\ \omega_{c_2} &= \frac{1}{\alpha\tau} = \omega_p \quad (\text{pole})\end{aligned}\right\}$$

$$\boxed{\omega_p > \omega_z} \checkmark$$

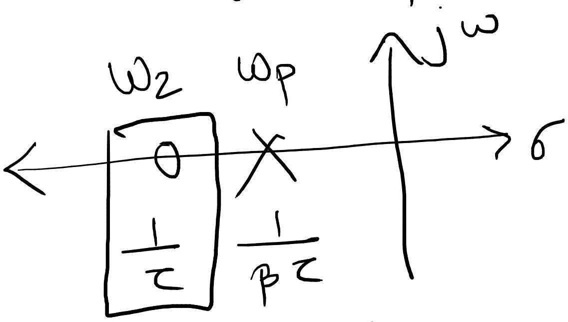


↑ Gain after certain freq



+ve change in phase
⇒ lead in phase
⇒ Lead-compensator

Lag compensator



$\boxed{\omega_z > \omega_p}$ ☆

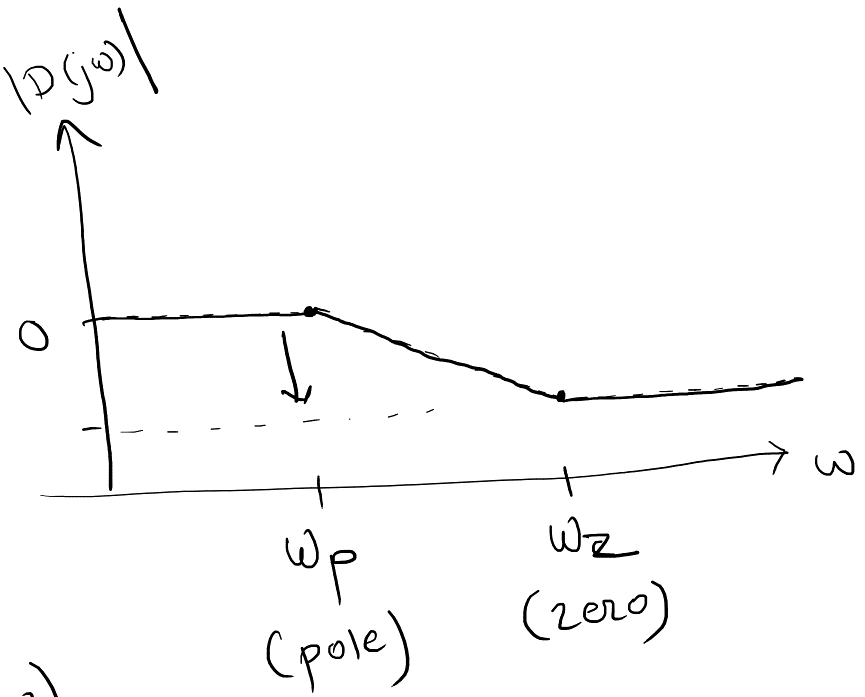
$$D(s) = \frac{s+\omega_z}{s+\omega_p} \times \frac{\omega_p}{\omega_z}$$

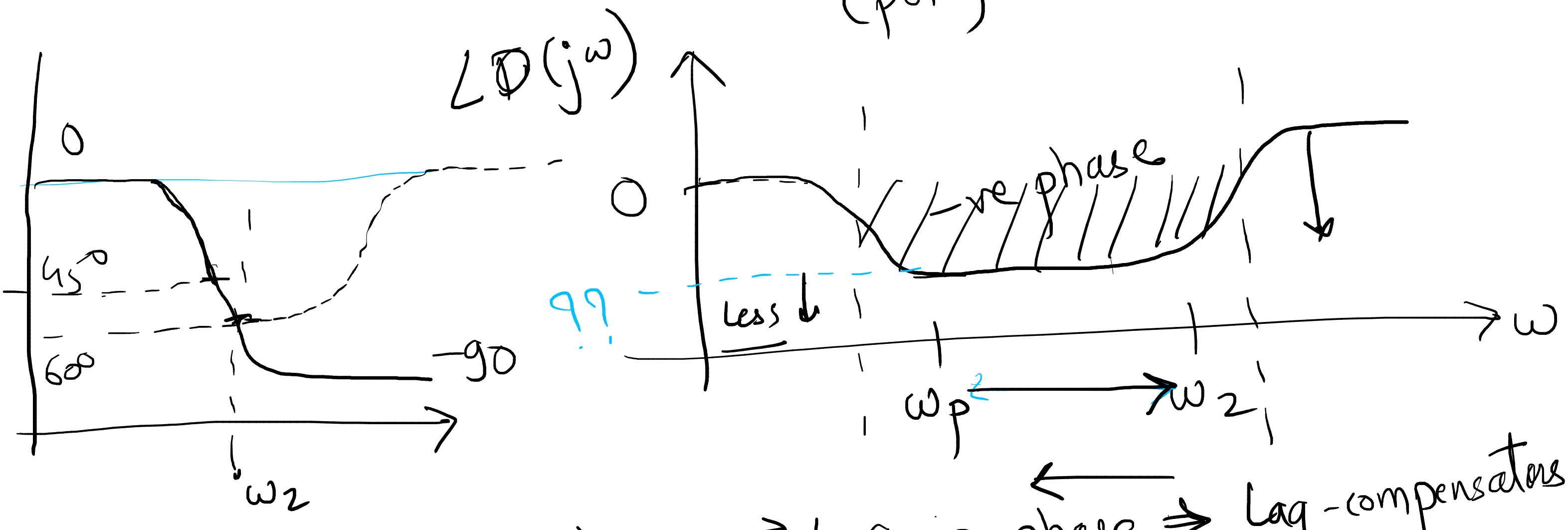
$$\boxed{D(s) = \frac{\tau s + 1}{\beta \tau s + 1}}$$

$\boxed{\beta > 1}$ $\boxed{\tau > 0}$

$$\omega_{c_1} = \omega_p = \frac{1}{\beta\tau}$$

$$\omega_{c_2} = \omega_z = \frac{1}{\tau}$$



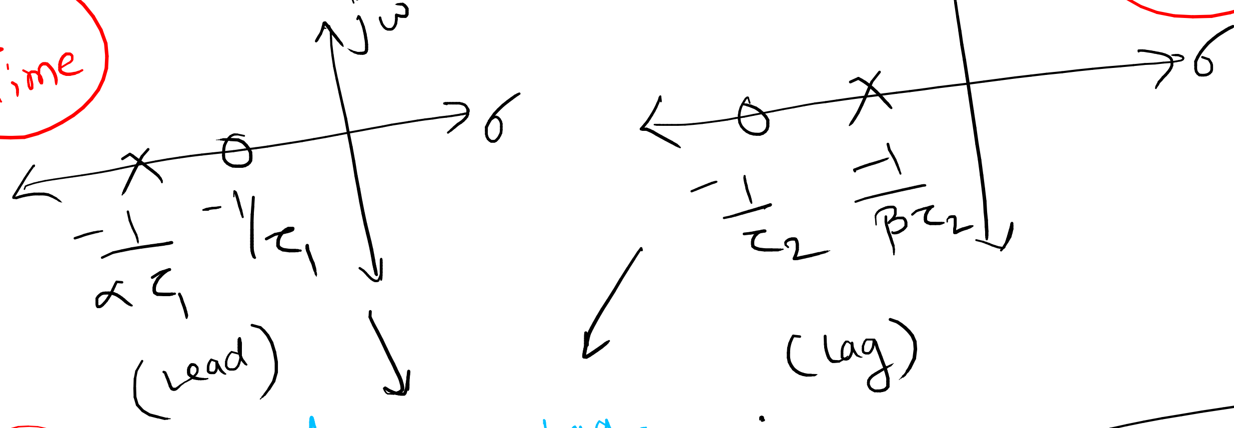


−ve phase ⇒ Lag in phase ⇒ Lag-compensators

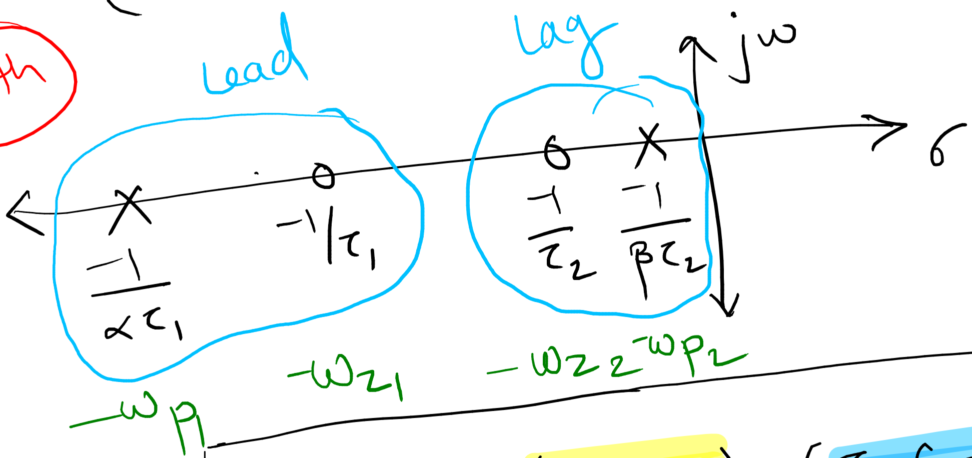
Lead-lag compensator

Time

Error



Both



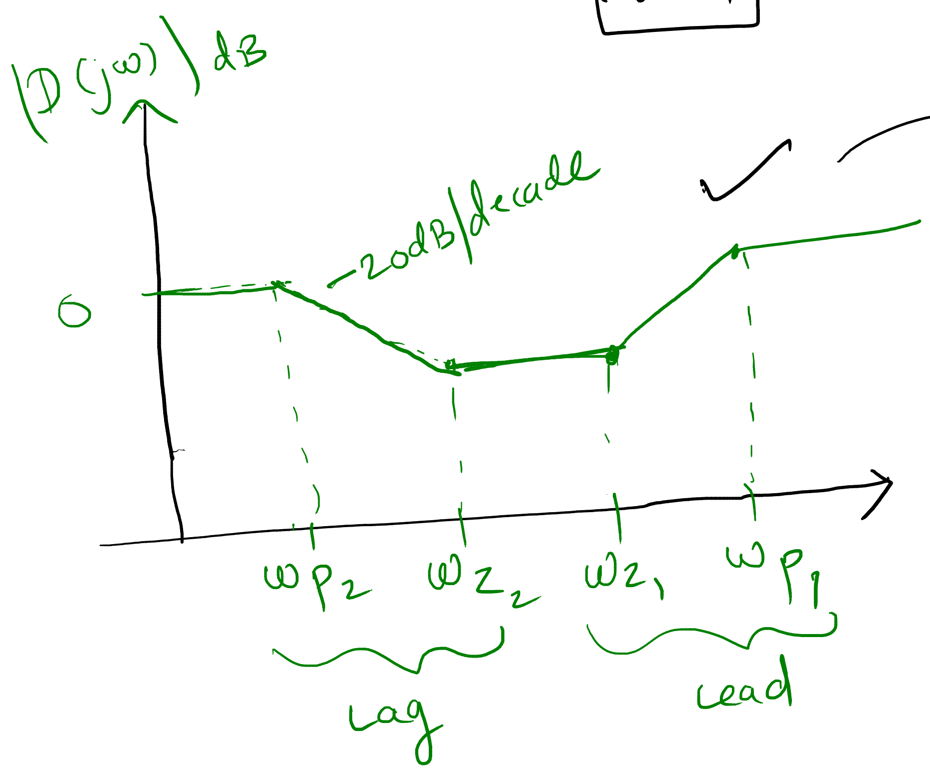
Lead-lag compensator

$\omega p_1 - \omega z_1$

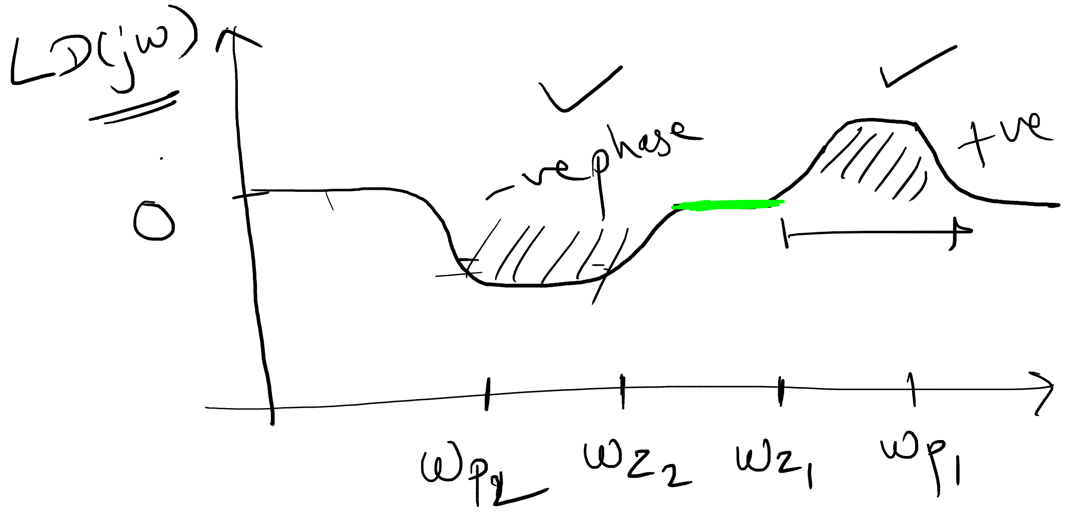
$-\omega z_2 - \omega p_2$

$$D(s) = \frac{(\tau_1 s+1)(\tau_2 s+1)}{(\alpha\tau_1 s+1)(\beta\tau_2 s+1)}$$

$\alpha < 1$ $\qquad$ $\beta > 1$

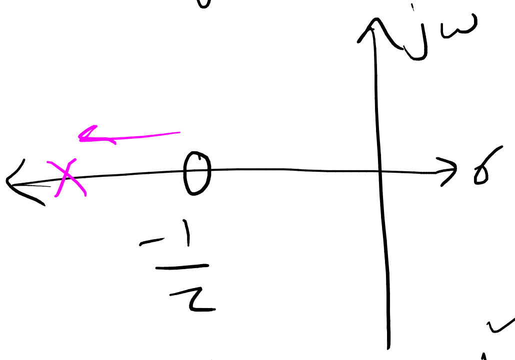


$G(s)$

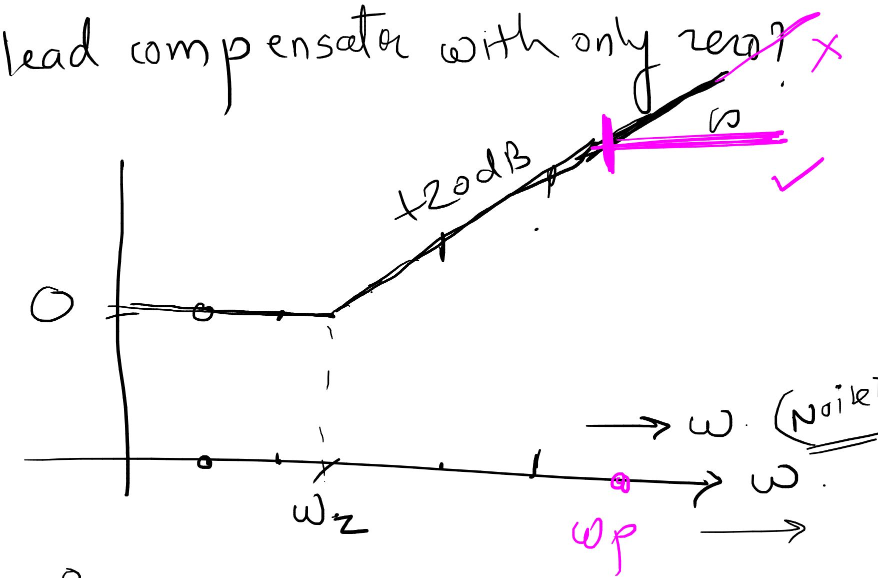


$\boxed{G(s)}$

★ why we don't have lead compensator with only zero?



$\Rightarrow \omega_z$

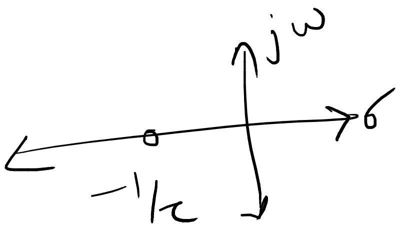


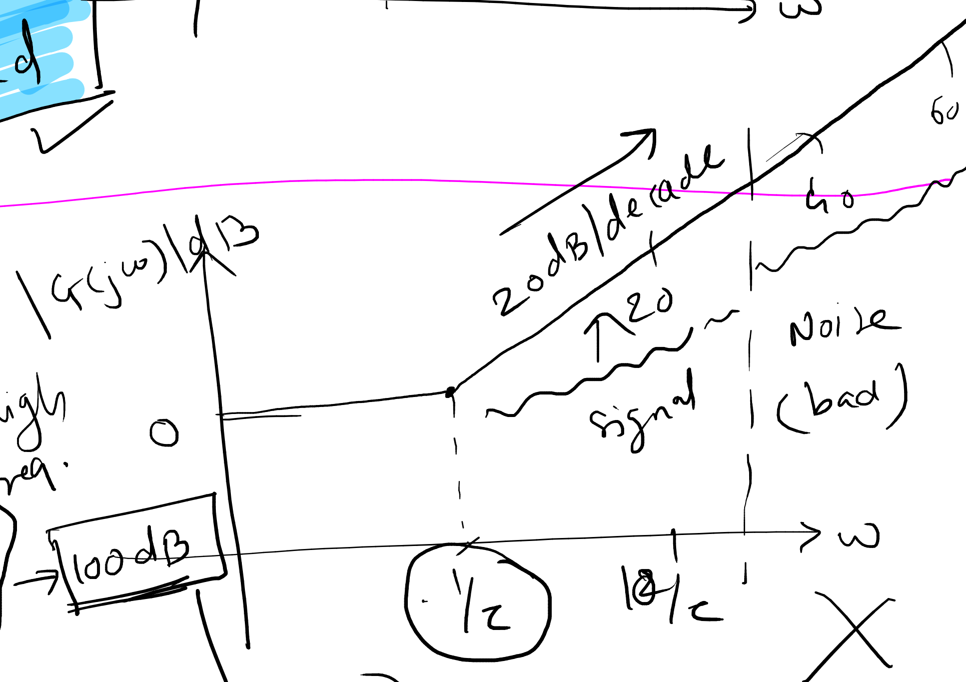
High ω ⇒ Very high gain

↳ Noise

Noise gets amplified

$\boxed{G(s) = \tau s + 1}$





Very high freq.

$\frac{10^6}{\tau} \rightarrow$ 100dB

very high gain

$\frac{10^4}{\tau} \rightarrow$ 80dB

$\frac{1000}{\tau} \rightarrow$ 60dB

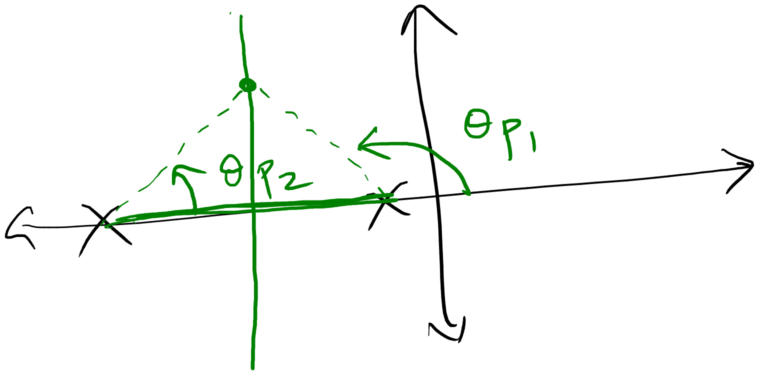
$\frac{100}{\tau} \rightarrow$ 40dB

Rule

1. For any point on RL

$$\sum(\angle \text{poles}) - \sum(\angle \text{zeros}) = 180^\circ$$

or multiple

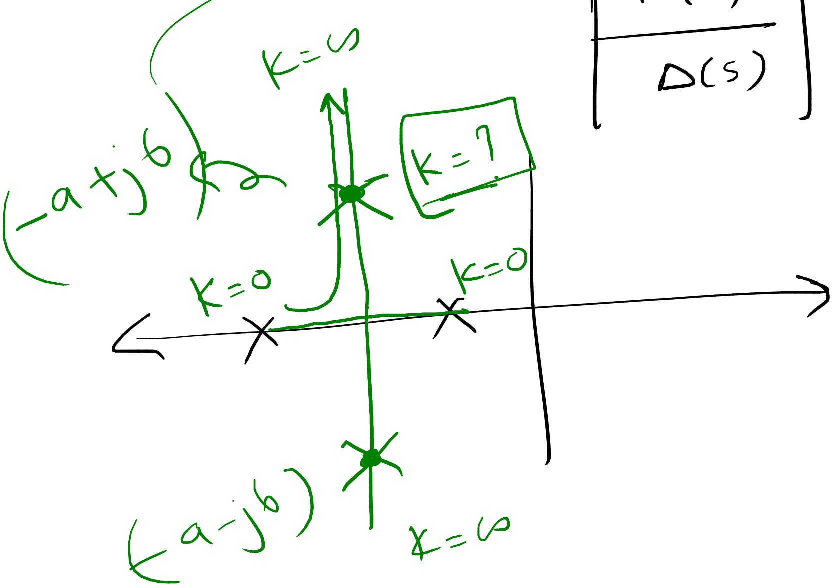


$$G(s) = \frac{K}{(s+1)(s+2)}$$

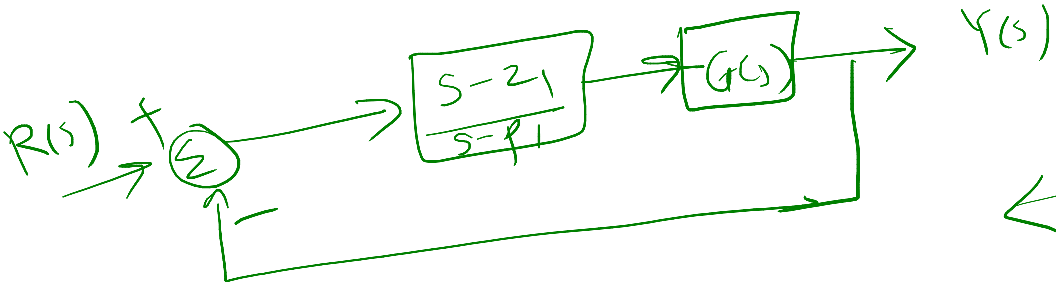
2. $$1 + K\frac{N(s)}{\Delta(s)} = 0$$

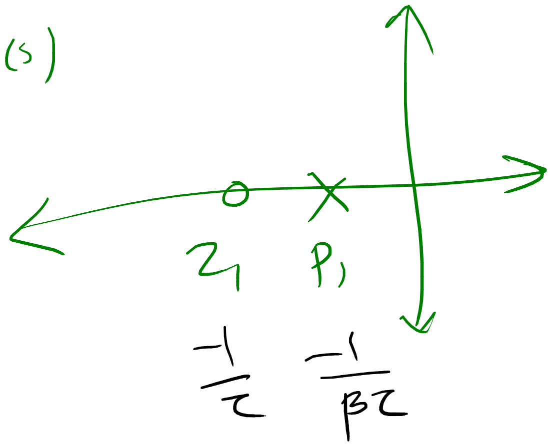
$$\left| K \frac{N(s)}{\Delta(s)} \right| = +1$$

$$\left| \frac{N(s)}{\Delta(s)} \right| = \frac{1}{|K|}$$



Lag-compensators : Minimize e_{ss}.





$$\boxed{G(s) = \frac{N(s)}{\Delta(s)}}$$

$$\boxed{\beta = \frac{z_1}{p_1} = \frac{\Delta(0) - \Delta(0)\cdot E_{ss_c}}{N(0)\, E_{ss_c}}}$$

(desired value)

$E_{ss_c} \rightarrow$ Expected error after compensation

$$e_{ss} = 0.05$$

$$\boxed{E_{ss_c} = 0.025}$$

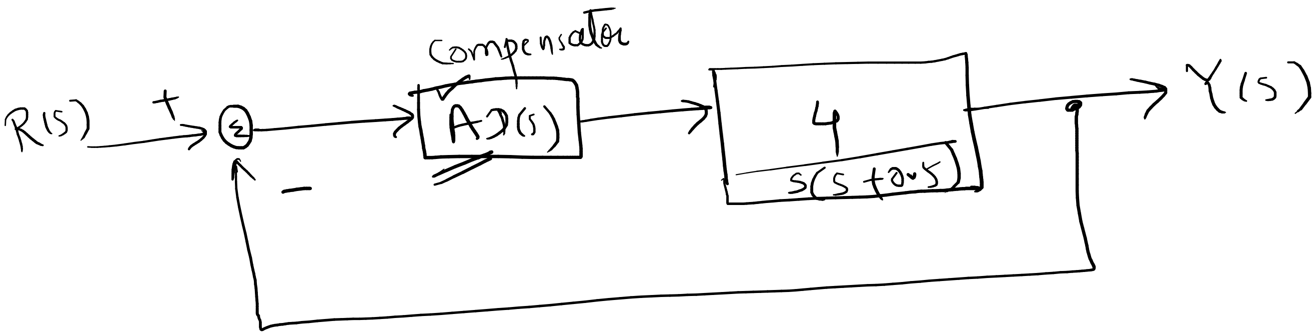
→ Desired value.

Example:

$$G(s) = \frac{4}{s(s+0.5)}$$

— Design a feedback system to meet the following specifications

1. Damping ratio of domninant closed-loop poles $\xi = 0.5$
2. Undamped natural frequency of domninant closed loop poles, $\omega_n = 5\ rad/s$
3. Velocity error constant = 80 sec^{-1}



Requirements → $\xi = 0.5$, $\omega_n = 5\ rad/s$, $K_v = 80\ sec^{-1}$

underdamped

$$G(s) = \frac{4}{s(s+0.5)}$$

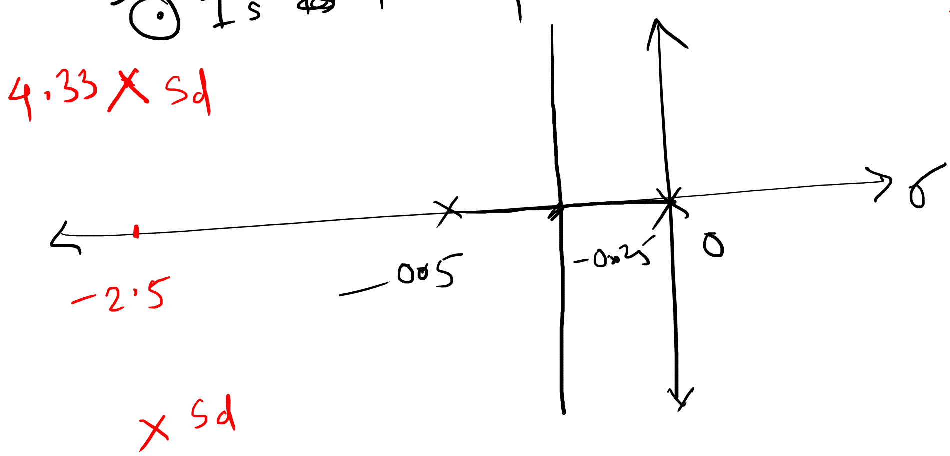
$$S_d = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

$$S_d = -(0.5 \times 5) \pm j\,5\sqrt{(1-0.5^2)}$$

$$\boxed{S_d = -2.5 \pm j4.33}$$ → Dominant pole or desired pole.

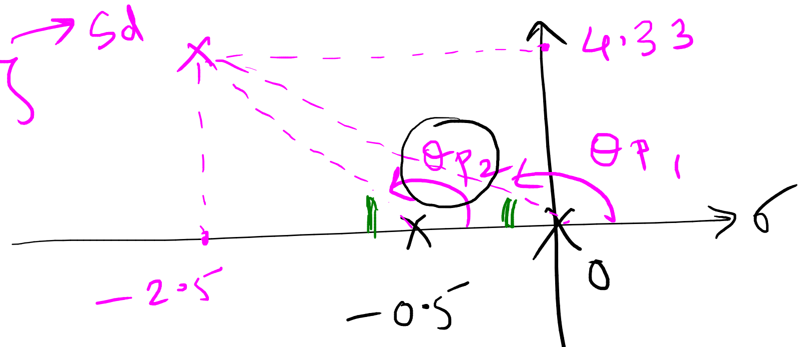
⊙ Is this point on the RL of G(s)?

→ NO



$\xi = 0.5$, $\omega_n = 5$ rad/s $\rightarrow S_d$

$$G(s) = \frac{4}{s(s+0.5)}$$



$$\Sigma(\angle \text{ with poles}) - \Sigma(\angle \text{ with zeros}) = 180^\circ$$

$$\angle S_d = (\theta p_1 + \theta p_2) - (0)$$

$\theta p_1 = 120^\circ$

$\theta p_2 = 115^\circ$

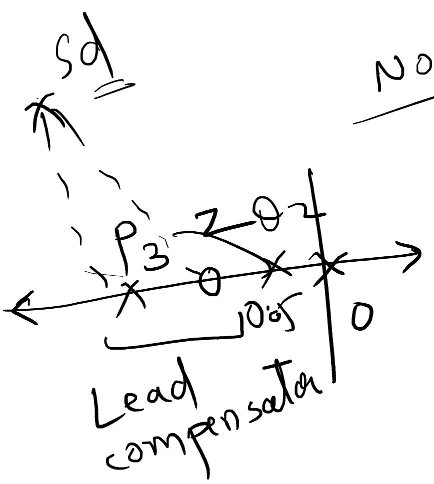
$$\angle S_d = 120 + 115 = \boxed{235^\circ}$$

Not on RL

Not 180 or multiple.

We need $180^\circ \leftarrow 235^\circ$ we have

Reduce $\angle$ by 55°

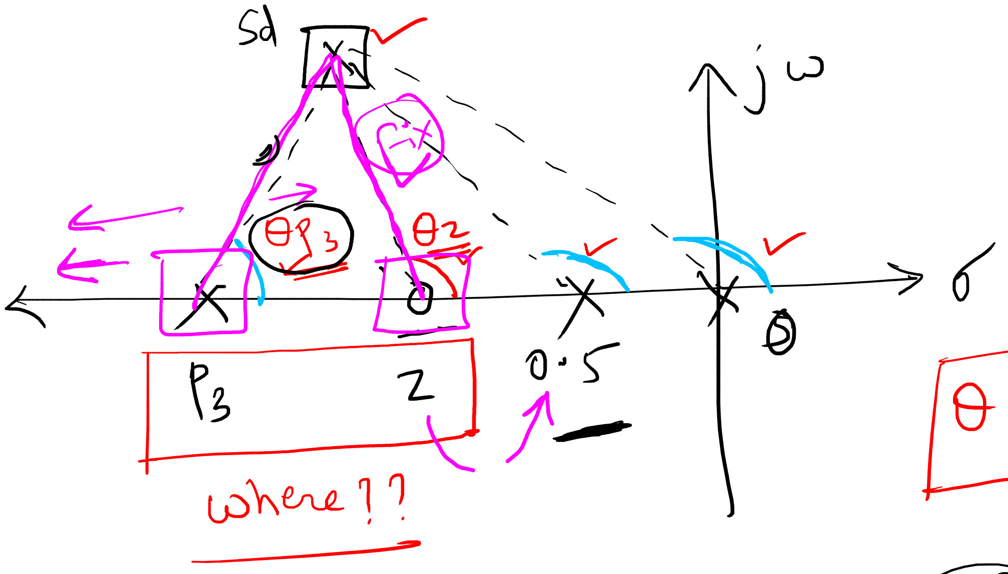


$$\phi = 235^\circ - 180 = 55^\circ \text{ (reduce)}$$

$$\angle S_d = (\theta p_1 + \theta p_2 + \theta p_3) - \theta z = 180^\circ$$

$$= 120 + 115 + \theta p_3 - \theta z = 180^\circ$$

$$\boxed{\theta p_3 - \theta z = -55^\circ}$$



$$\frac{4}{s(s+0.5)}$$

$$\boxed{\theta_{p_3} - \theta_z = -55^\circ}$$

$$D_1(s) = \frac{s + 1/\tau_1}{s + \frac{1}{\alpha \tau_1}} \qquad \alpha < 1$$

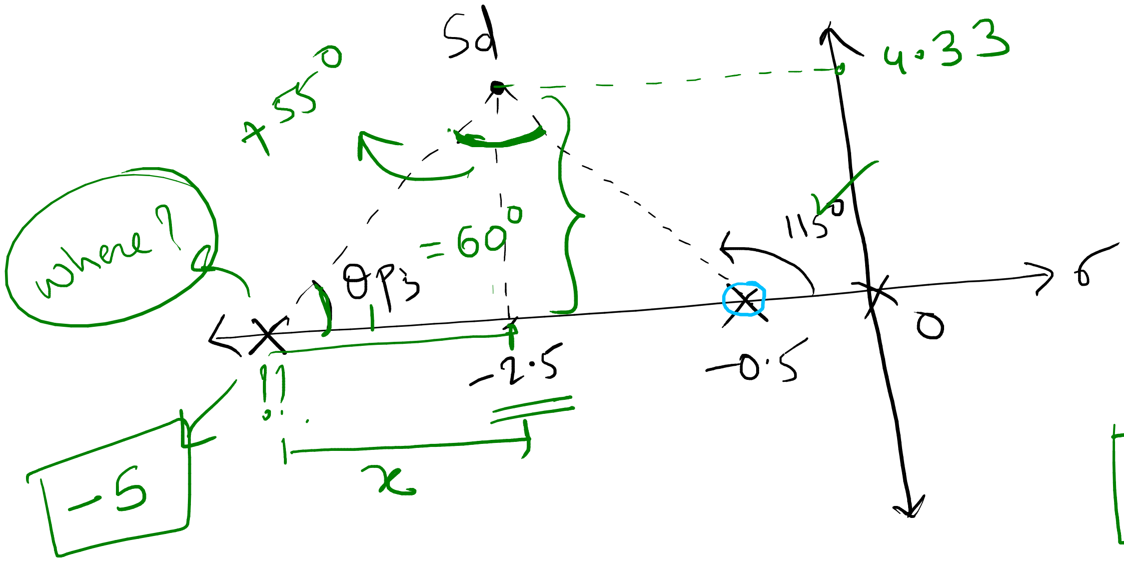
→ zero should close to the nearest pole of G(s).

Assume zero at $\boxed{s = -0.5}$

$$\theta_{p_3} - \theta_z = -55$$

$$\boxed{\theta_z = 115^\circ}$$

$$\theta_{p_3} = \boxed{-55} + \theta_z = 60^\circ$$

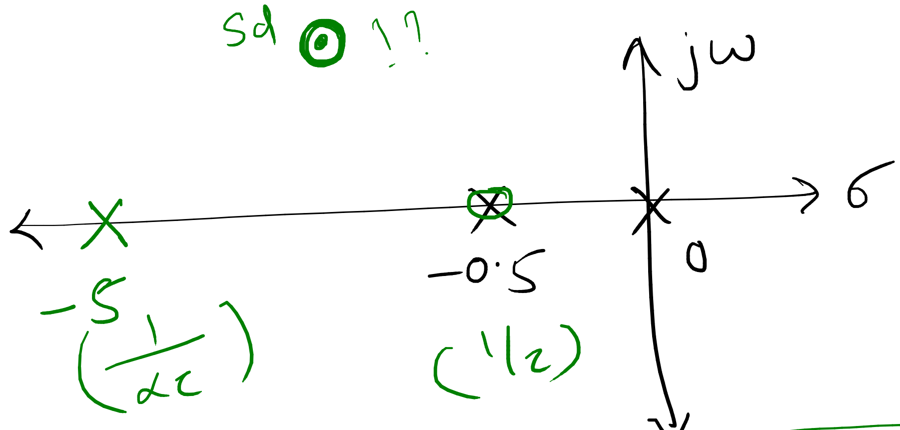


$$\tan 60 = \frac{4.33}{x}$$

$$x = \frac{4.33}{\tan 60}$$

$$\boxed{x = \sim 2.5}$$

sd ??



$$G(s) = \frac{4}{s(s+0.5)}$$

$$\frac{1}{\tau} = 0.5$$

$$\frac{1}{\alpha\tau} = 5$$

$$\boxed{\alpha = 0.1}$$

Lead compensator

$$\boxed{D_1(s) = \frac{s+0.5}{s+5} \; ; \; \alpha = 0.1}$$

Find K such that $\underline{s_d = -2.5 \pm 4.33\,j}$

$$D_1(s)G(s) = \frac{A\cancel{(s+0.5)}}{(s+5)} \times \frac{4}{s\cancel{(s+0.5)}} \; \star$$

$\star$ $\boxed{K = 4A}$

$$|D_1(s) \cdot G(s)| = \frac{K(1)}{s(s+5)}$$

OLTF

$$K = \left| s(s+5) \right|_{s_d}$$

$$\boxed{K = 25}$$

$$4A = K \Rightarrow \boxed{A = 6.25}$$

Lag compensator : $K_v = 80 \text{ sec}^{-1}$

For current system $K_v = \lim_{s \to 0} s \cdot \underbrace{D_1(s)}_{\text{lead}} \cdot \underbrace{G(s)}_{\text{plant}}$

$$= \lim_{s \to 0} \cancel{s} \cdot \left[\frac{K}{\cancel{s}(s+5)} \right]$$

$$= \frac{K}{5} = \frac{25}{5} = 5$$

For new system $K_v = 80$ (requirement)

$$e_{ss} = \frac{1}{K_v} \quad K_v \uparrow \quad e_{ss} \downarrow .$$

Lag compensator $D_2(s) = \dfrac{s + 1/\tau_2}{s + \dfrac{1}{\beta \tau_2}}$ $\beta > 1$

Typically we choose $\beta = 1/\alpha$.

But $\alpha = 0.1$ $\therefore$ $\boxed{\beta = 10}$

with general thumb rule

Error constant $\times \beta$

$K_v = 5 \times 10 = 50$

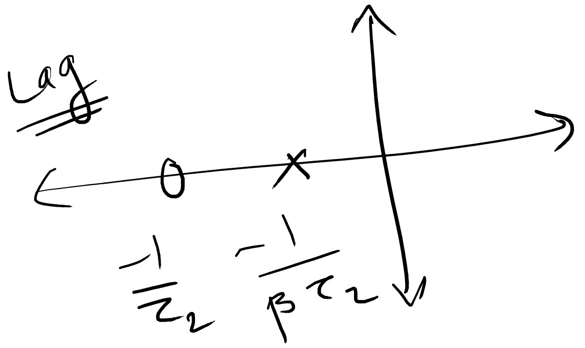
But we want $K_v = 80$

calculated above ← $K_v = 5$ → without lag comp.

$K_v^d = 80$ → desired value.

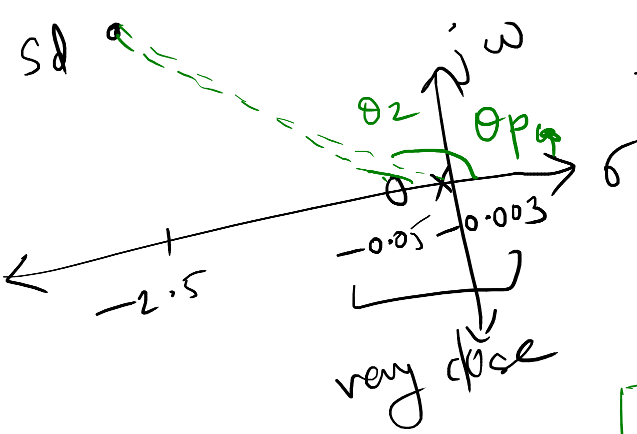
$$\beta = \frac{K_v^d}{K_v} = \frac{80}{5} = \underline{16}.$$

Lag



$$\boxed{\frac{1}{\tau_2} = \frac{Re(S_d)}{50}}$$ General rule

50 → minimum 50.



$$\frac{1}{\tau_2} = \frac{-2.5}{50} = -0.05$$ (zero)

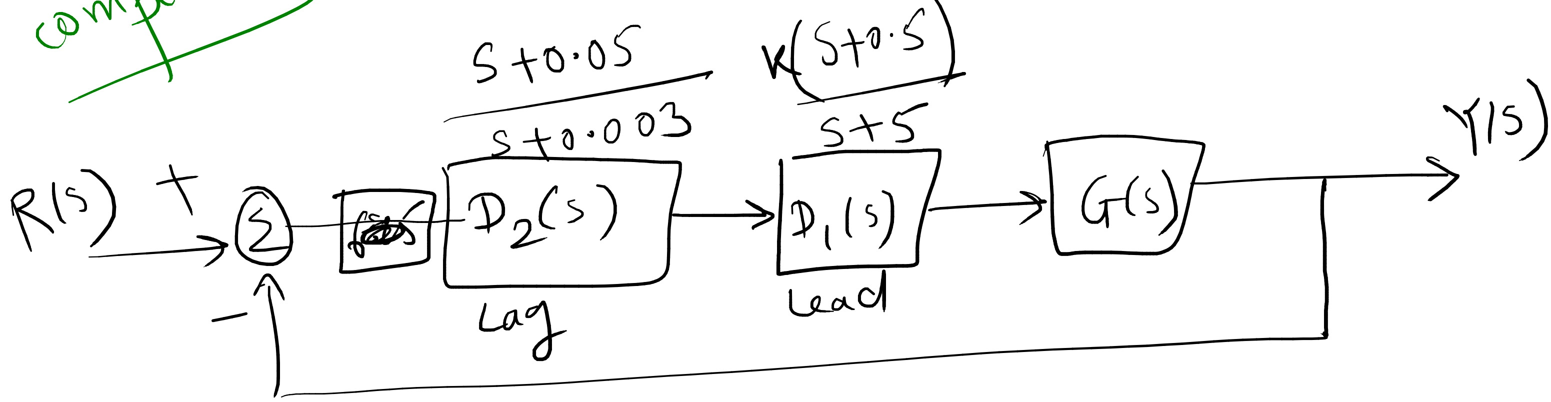
$$\frac{1}{\beta\tau_2} = \frac{-0.05}{16} = -0.003$$ (pole)

$$\boxed{\theta_p - \theta_2 = 0}$$

$$\boxed{\theta_p \approx \theta_2}$$

↳ Lead comp. design is not affected

Lag compensator, $D_2(s) = \frac{s+0.05}{s+0.003}$



$$D_2(s) \cdot D_1(s) = \underbrace{\frac{25}{4}}_{\text{(Gain)}} \cdot \underbrace{\frac{(s+0.5)}{(s+5)}}_{\text{(Lead)}} \underbrace{\left(\frac{s+0.05}{s+0.003}\right)}_{\text{(Lag)}} \underbrace{\frac{4}{s(s+0.5)}}_{(G(s))}$$

Lead ↓

$\xi = 0.5$

$\omega_n = 5\ rad/s$

Lag ↓

$K_v = 80\ sec^{-1}$